Pratham Aggarwal

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Education

Sep 2024–Mar 2027 **University of California San Diego**, La Jolla, CA

B.S. in Data Science

Honors: HDSI Scholar; HKN-IEEE.

Research Interests: Machine Learning, Climate Science, Probability & Statistics.

Experience

Jun 2025–Now **Undergraduate Researcher**, *Climate Analytics Lab*, UC San Diego, La Jolla, CA
Advisor: Duncan Watson-Parris
- Developed FASST, a lightweight coupled ML climate model combining a non-autoregressive U-Net atmospheric model, the POP2 ocean model, and a sea-ice model.
- Demonstrated stable climate behavior in a 140-year unforced simulation with realistic interannual ENSO variability.
- Developed a diffusion-based atmospheric model to predict 6-hourly fields, including wind, pressure, precipitation, etc., from sea-surface temperature and solar insolation.
- Contributed to ClimateBench 2.0, evaluating historical and future climate predictions from 14 CMIP6 climate models using statistical metrics; awarded a $4,500 HDSI scholarship.

Apr–Sep 2026 **Undergraduate Researcher**, *Belkin Lab*, UC San Diego, La Jolla, CA
Advisor: Qingsong Wang
- Investigated steering techniques to obtain topic specific refusal in diffusion based LLMs.
- Applied layer-wise probing to identify topic-specific information, found that self-harm and general refusal are encoded along nearly orthogonal directions.

Aug–Dec 2025 **Undergraduate Researcher**, *Existential Robotics Lab*, UC San Diego, La Jolla, CA
Advisor: Nikolay A. Atanasov
- Evaluated A*, RRT, and MPC motion-planning algorithms in dynamic environments, reducing path-planning failures by 28% while maintaining average trajectory error below 0.15 m.
- Increased simulation speed by 2.1 times by parallelizing physics evaluations and testing motion-planning algorithms concurrently.

Jan–Apr 2026 **AI/ML Research Intern**, *NIOV Labs*, San Diego, CA
Advisor: Sadeil Lewis
- Reduced API costs by 23% by developing a task-routing system that used prompt-based classification to assign requests to AI agents based on task complexity.
- Developed rubric-guided prompting to evaluate LLM responses for tone, length, and content alignment with target requirements.

Mar–Jun 2025 **Quantitative Developer Intern**, *SFIC Quantitative Technologies*, San Diego, CA
- Increased simulated options trading returns by 12% for a $1.3M student-run investment fund by building a deep learning model to select option strategies and exercise timing.

Jan–Mar 2025 **Data Science Intern**, *Solana Center*, San Diego, CA
- Led a 5-member team in analyzing composting records using SQL
- Developed a real-time Power BI dashboard to track key metrics.
- Developed automated Python pipelines for data cleaning and missing-value imputation, improving the speed, consistency, and reproducibility of data preparation.

Publications

NeurIPS 2026 **P. Aggarwal**, D. Watson-Parris, and W. E. Chapman. FASST: A Compact Stateless Atmosphere Model for Long-Term Ocean Coupling. In *Tackling Climate Change with Machine Learning Workshop at NeurIPS*, 2026. arXiv:2610.36297.

Presentations

HDSI 2025 **P. Aggarwal**, K. Shah, W. Tobin, and D. Watson-Parris. ClimateBench 2.0: Probabilistic Climate Model Scoring. Poster presented at the Halıcıoğlu Data Science Institute (HDSI), UC San Diego, 2025. Web app: climate-analytics-lab.github.io/ClimateBench_app.

Projects

Coastal Flooding Prediction Under Out-of-Distribution Conditions, *NSF Hackathon, 1st Place*
- Improved coastal flood prediction F1-score from 0.53 to 0.94 through 30+ model experiments, selecting CatBoost as the final model.
- Improved model performance through iterative feature engineering, incorporating features based on coastal flooding physics during a 48-hour hackathon.

Homeless Services Geospatial Platform, *Brain Corp Hackfrontier, 1st Place*
- Developed a geospatial site-selection model using 35+ transit and geographic features to identify optimal homeless service center locations, achieving 67% placement accuracy.
- Built a real-time computer vision system to track geographic shifts in homeless populations and inform service center placement.

Skills

Languages: Python, SQL, C, C++, Java

AI / ML: PyTorch, TensorFlow, Scikit-Learn, Pandas, Matplotlib, Power BI, Tableau

Systems & Tools: Docker, Kubernetes, AWS, Google Cloud, Git/GitHub, HPC (NCAR Derecho, Casper; NSF NRP)